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ECON1050 Tutorial 9

Integration

Week 11 - Semester 1 2026

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Today's Plan

- Recap: key ideas from Lecture 9
- Tutorial questions: Q1–Q4

Recap: What is integration?

Core idea

Integration **undoes differentiation**.

Indefinite integral

$$\int f(x)\,dx = F(x) + C, \qquad F'(x) = f(x).$$

- $F(x)$ is an **antiderivative** of $f(x)$.
- C is the **constant of integration**.
- C can be any scalar, so the integral is called **indefinite**.

Economic interpretation

Integrating a marginal function can recover a total function up to a constant.

Recap: Rules for indefinite integrals I

Power rule

For $a \neq -1$,

$$\int x^a \, dx = \frac{x^{a+1}}{a+1} + C.$$

Constant and scalar rules

$$\int k \, dx = kx + C, \qquad \int kf(x) \, dx = k \int f(x) \, dx.$$

Sum rule

$$\int [f(x) \pm g(x)] \, dx = \int f(x) \, dx \pm \int g(x) \, dx.$$

Recap: Rules for indefinite integrals II

Logarithmic rules

$$\int \frac{1}{x}\, dx = \ln|x| + C, \qquad \int \ln x \, dx = x \ln x - x + C.$$

Exponential rules

$$\int e^x \, dx = e^x + C, \qquad \int e^{kx} \, dx = \frac{1}{k} e^{kx} + C.$$

$$\int a^x \, dx = \frac{a^x}{\ln a} + C.$$

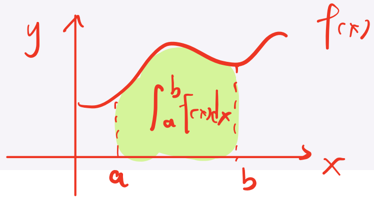
Recap: Definite integrals

Definition

A definite integral is a number:

$$\int_a^b f(x)\,dx.$$

It represents the **net area** under the curve between a and b.



Fundamental Theorem of Integral Calculus

If $F'(x) = f(x)$, then

antiderivative

$$\int_a^b f(x)\,dx = F(b) - F(a) = \Big[F(x)\Big]_{x=a}^{x=b}$$

Net area vs total area

Areas above the x-axis count positively; areas below the x-axis count negatively.

Recap: Properties of definite integrals

Splitting an interval

For $a < c < b$,

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx.$$

Changing limits

$$\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx, \qquad \int_a^a f(x)\,dx = 0.$$

Constant multiple

$$\int_a^b \alpha f(x)\,dx = \alpha \int_a^b f(x)\,dx.$$

Recap: Integration by parts

Integration by parts

If $f(x)$ and $g(x)$ are differentiable, then

$$\int f(x)g'(x)\,dx = f(x)g(x) - \int f'(x)g(x)\,dx.$$

When to use it

Use integration by parts for products where the basic rules or substitution do not work directly.

Common strategy

Choose one factor to differentiate and the other factor to integrate so that the remaining integral becomes simpler.

Recap: Integration by substitution

Integration by substitution

If

$$u = g(x), \qquad du = g'(x)\,dx,$$

then

$$\int \underbrace{f(g(x))}_{f(u)} \underbrace{g'(x)\,dx}_{du} = \boxed{\int f(u)\,du.}$$

$u = g(x)$

$\frac{du}{dx} = g'(x)$

$du = g'(x) \cdot dx$

Common substitution patterns

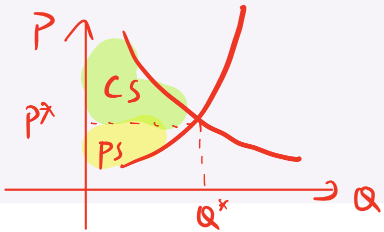
- **Power substitution**: one term is the derivative of another.
- **Exponential substitution**: one term is exponential in a function of x.
- **Logarithmic substitution**: numerator is the derivative of denominator.

Recap: Welfare applications

Consumer surplus

If inverse demand is $D(Q)$ and equilibrium is (P^*, Q^*), then

$$CS = \int_0^{Q^*} D(Q)\,dQ - P^* Q^*.$$



Producer surplus

If inverse supply is $S(Q)$ and equilibrium is (P^*, Q^*), then

$$PS = P^* Q^* - \int_0^{Q^*} S(Q)\,dQ.$$

Interpretation

Surplus measures gains relative to the equilibrium market price.

Tutorial Q1

Question 1

Evaluate:

indefinite

$$(1) \quad \int (5x^3 + 2)\, dx$$

$$(2) \quad \int \left(2x^2 + \frac{1}{x^3}\right) dx$$

$+ C$

$C_1 + C_2 = C$

define

$$(3) \quad \int_0^2 4x(x^2+1)\, dx \Rightarrow \text{number}$$

(1) $\int (5x^3+2)dx = \int 5x^3 dx + \int 2 dx = 5\int x^3 dx + 2\int dx = \frac{5}{4}x^4 + C_1 + 2x + C_2 = \frac{5}{4}x^4 + 2x + C$

(2) $\int (2x^2 + \frac{1}{x^3})dx = 2\int x^2 dx + \int x^{-3} dx = \frac{2}{3}x^3 + C_1 - \frac{1}{2}x^{-2} + C_2 = \frac{2}{3}x^3 - \frac{1}{2}\frac{1}{x^2} + C$

(3) $\int_0^2 4x(x^2+1)dx = \int_0^2 (4x^3+4x)dx = 4\int_0^2 x^3 dx + 4\int_0^2 x\, dx$

$= 4\left[\frac{x^4}{4}\right]_{x=0}^{x=2} + 4\cdot\left[\frac{x^2}{2}\right]_{x=0}^{x=2}$

$= \left[x^4\right]_{x=0}^{x=2} + 2\left[x^2\right]_{x=0}^{x=2} = (2^4 - 0^4) + 2(2^2 - 0^2) = 24$

Tutorial Q1: Basic integration

Knowledge used

- Use the power rule for indefinite integrals.
- Use linearity to integrate term by term.
- For definite integrals, apply $F(b) - F(a)$.

Solution

$$\int (5x^3 + 2)\,dx = 5\int x^3\,dx + \int 2\,dx = \frac{5}{4}x^4 + 2x + C.$$

$$\int \left(2x^2 + \frac{1}{x^3}\right)\,dx = 2\int x^2\,dx + \int x^{-3}\,dx = \frac{2}{3}x^3 - \frac{1}{2}x^{-2} + C.$$

$$\int_0^2 4x(x^2+1)\,dx = \int_0^2 (4x^3 + 4x)\,dx = \left[x^4 + 2x^2\right]_0^2 = 24.$$

Tutorial Q2

Integration by parts

If $f(x)$ and $g(x)$ are differentiable, then

$$\int f(x)g'(x)\,dx = f(x)g(x) - \int f'(x)g(x)\,dx.$$

Question 2

Evaluate using integration by parts:

$$(1) \quad \int \frac{\ln x}{x^2}\,dx, \qquad x > 0.$$

5 mins

$$(2) \quad \int_1^2 4xe^{2x}\,dx.$$

(1) $\frac{\ln x}{x^2}$: $f(x) = \ln x \Rightarrow f'(x) = \frac{1}{x}$

$g'(x) = x^{-2} \Rightarrow g(x) = -x^{-1}$

$\int \frac{\ln x}{x^2}dx = \int f(x)\cdot g'(x)\,dx = f(x)\cdot g(x) - \int f'(x)\cdot g(x)\,dx = \frac{-\ln x}{x} - \int \frac{1}{x}\cdot\left(\frac{-1}{x}\right)dx = \frac{-\ln x}{x} + \int \frac{1}{x^2}dx$

$= -\frac{\ln x}{x} - \frac{1}{x} + C$

(2) $4x\cdot e^{2x}$: $f(x) = 4x \Rightarrow f'(x) = 4$

$g'(x) = e^{2x} \Rightarrow g(x) = \frac{1}{2}\cdot e^{2x}$

$\int_1^2 4xe^{2x}dx = \left[4x\cdot\frac{1}{2}\cdot e^{2x}\right]_{x=1}^{x=2} - \int_1^2 4\cdot\frac{1}{2}\cdot e^{2x}dx = \left[2xe^{2x}\right]_{x=1}^{x=2} - \left[e^{2x}\right]_{x=1}^{x=2}$

$= (2\cdot 2\cdot e^{2\cdot 2} - 2\cdot 1\cdot e^{2}) - (e^{2\cdot 2} - e^{2}) = 3e^4 - e^2$

Tutorial Q2(1): Integration by parts

Knowledge used

- Use the integration by parts formula

$$\int f(x)g'(x)\,dx = f(x)g(x) - \int f'(x)g(x)\,dx.$$

- Choose $f(x) = \ln x$ because it becomes simpler after differentiation.

Solution

Let

$$f(x) = \ln x, \qquad f'(x) = \frac{1}{x}, \qquad g'(x) = x^{-2}, \qquad g(x) = -\frac{1}{x}.$$

Then

$$\int \frac{\ln x}{x^2}\,dx = \ln x\left(-\frac{1}{x}\right) - \int \frac{1}{x}\left(-\frac{1}{x}\right)\,dx.$$

$$= -\frac{\ln x}{x} + \int x^{-2}\,dx = -\frac{\ln x}{x} - \frac{1}{x} + C = -\frac{\ln x + 1}{x} + C.$$

Tutorial Q2(2): Definite integral by parts

Knowledge used

- Use integration by parts for the product $4xe^{2x}$.
- Evaluate the antiderivative at the upper and lower bounds.

Solution

Let

$$f(x) = 4x, \qquad f'(x) = 4, \qquad g'(x) = e^{2x}, \qquad g(x) = \frac{1}{2}e^{2x}.$$

Then

$$\int_1^2 4xe^{2x}\,dx = \left[2xe^{2x}\right]_1^2 - 2\int_1^2 e^{2x}\,dx.$$

$$= \left[2xe^{2x}\right]_1^2 - \left[e^{2x}\right]_1^2 = (4e^4 - 2e^2) - (e^4 - e^2).$$

$$= 3e^4 - e^2 \approx 156.4.$$

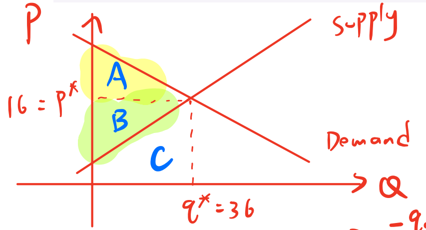
Question 3

Demand and supply are

$$q_d = 100 - 4p, \qquad q_s = -12 + 3p.$$

5 mins

Find the equilibrium price and quantity. Then compute consumer surplus and producer surplus.



$$CS = A = (A+B+C) - (B+C) = \int_0^{q^*} \text{Demand}\, dq - p^* \cdot q^*$$

$$PS = B = (B+C) - C = p^* \cdot q^* - \int_0^{q^*} \text{supply}\, dq$$

$$\begin{cases} q_d = 100 - 4p & (1) \Leftrightarrow p = \dfrac{-q_d + 100}{4} \\ q_s = -12 + 3p & (2) \Leftrightarrow p = \dfrac{q_s + 12}{3} \end{cases}$$

$(2) - (1)$: $0 = (-12 + 3p) - (100 - 4p)$

$p^* = 16$

From (1): $q^* = 100 - 4p^* = 36$

$$CS = \int_0^{36} \frac{-q + 100}{4}\, dq - 16 \times 36$$

$$= \left[25q - \frac{1}{8}q^2\right]_{q=0}^{q=36} - 576 = \left[25 \times 36 - \frac{1}{8} \times 36 - 0\right] - 576 = 162$$

$$PS = 576 - \int_0^{36} \frac{q + 12}{3}\, dq$$

$$= 576 - \left[4 \times 36 + \frac{1}{6} \times 36^2 - 0\right]$$

$$= 216$$

Tutorial Q3: Equilibrium

Knowledge used

- Convert demand and supply into inverse demand and inverse supply.
- Equilibrium occurs where inverse demand equals inverse supply.

Solution

Inverse demand:

$$q_d = 100 - 4p \Rightarrow p_d = 25 - \frac{1}{4}q.$$

Inverse supply:

$$q_s = -12 + 3p \Rightarrow p_s = 4 + \frac{1}{3}q.$$

At equilibrium:

$$25 - \frac{1}{4}q = 4 + \frac{1}{3}q \Rightarrow \frac{7}{12}q = 21 \Rightarrow q^* = 36.$$

Then

$$p^* = 4 + \frac{1}{3}(36) = 16.$$

Tutorial Q3: Consumer and producer surplus

Knowledge used

- Consumer surplus is the area under inverse demand above equilibrium price.
- Producer surplus is the area above inverse supply below equilibrium price.

Solution

$$CS = \int_0^{36} \left(25 - \frac{1}{4}q\right) dq - 16(36).$$

$$= \left[25q - \frac{1}{8}q^2\right]_0^{36} - 576 = 25(36) - \frac{1}{8}(36)^2 - 576 = 162.$$

$$PS = 16(36) - \int_0^{36} \left(4 + \frac{1}{3}q\right) dq.$$

$$= 576 - \left[4q + \frac{1}{6}q^2\right]_0^{36} = 576 - 4(36) - \frac{1}{6}(36)^2 = 216.$$

Tutorial Q4

Integration by parts

If $f(x)$ and $g(x)$ are differentiable, then

$$\int f(x)g'(x)\,dx = f(x)g(x) - \int f'(x)g(x)\,dx.$$

Question 4

Evaluate:

$$\int \ln x\,dx.$$

Hint: integration by part

3 min

$$\int \ln x\,dx = \int 1 \cdot \ln x\,dx$$

$$\begin{cases} f(x) = \ln x \Rightarrow f'(x) = \frac{1}{x} \\ g'(x) = 1 \Rightarrow g(x) = x \end{cases}$$

$$\int \ln x\,dx = \ln x \cdot x - \int \frac{1}{x} \cdot x\,dx$$

$$= x \cdot \ln x - \int dx = x\ln x - (x + c)$$

$$= x\ln x - x - c$$

Tutorial Q4

Knowledge used

- Treat $\ln x$ as $\ln x \cdot 1$.
- Use integration by parts.

Solution

Let

$$f(x) = \ln x, \qquad f'(x) = \frac{1}{x}, \qquad g'(x) = 1, \qquad g(x) = x.$$

Then

$$\int \ln x \, dx = x \ln x - \int \frac{1}{x} \cdot x \, dx = x \ln x - \int 1 \, dx.$$

Therefore,

$$\int \ln x \, dx = x \ln x - x + C.$$

Key takeaways

- Indefinite integration recovers antiderivatives; definite integration gives net area.
- Integration by parts is the reverse of the product rule.
- Integration by substitution is the reverse of the chain rule.
- Consumer and producer surplus are areas calculated using definite integrals.